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CASE STUDY 1: UNIX, LINUX, AND ANDROID

In the previous chapters, we took a close look at many operating system principles, abstractions, algorithms, and techniques in general. Now it is time to look at some concrete systems to see how these principles are applied in the real world. We will begin with Linux, a popular variant of UNIX, which runs on a wide variety of computers. It is the dominant operating systems on high-end workstations and servers, but it is also used on systems ranging from smartphones (Android is based on Linux) to supercomputers.

Our discussion will start with its history and evolution of UNIX and Linux. Then we will provide an overview of Linux, to give an idea of how it is used. This overview will be of special value to readers familiar only with Windows, since the latter hides virtually all the details of the system from its users. Although graphical interfaces may be easy for beginners, they provide little flexibility and no insight into how the system works.

Next, we come to the heart of this chapter, an examination of processes, memory management, I/O, the file system, and security in Linux. For each topic, we will first discuss the fundamental concepts, then the system calls, and finally the implementation.

Right off the bat we should address the question: Why Linux? Linux is a variant of UNIX, but there are many other versions and variants of UNIX including AIX, FreeBSD, HP-UX, SCO UNIX, System V, Solaris, and others. Fortunately, the fundamental principles and system calls are pretty much the same for all of them (by design). Furthermore, the general implementation strategies, algorithms,